

Pre-Feasibility Study Medical Olive Oil Soap Factory



2017

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1. Executive Summary

The idea of the project is to build a factory in Jarash governorate to manufacture medical soap of different shapes and sizes using olive oil. The idea was adopted due to the high demand for this type of soap in Jordan, the availability of raw materials and local expertise, and export markets (such as United States of America, Europe, and Arab countries).

Payback period is 5 years with an internal rate of return of 21.98%. Moreover, the project will create 15 job opportunities.

The following table shows the most important results of the study.

Table1 main results of the study

Results	
Total investment cost (JD)	596,320
IRR (%)	%22.34
Payback period (years)	5
Net present value (for 10 years)	338,647
Created job opportunities	15
Expected total cash inflows (for 10 years)	4,659,468
Total operating expenses (for 10 years)	2,403,512
Total net profit (for 10 years)	695,325

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

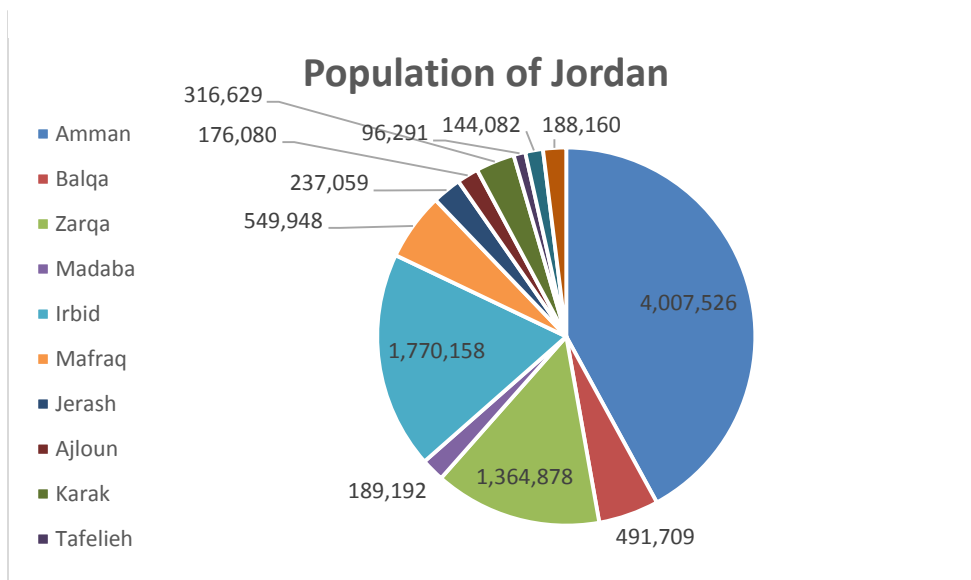
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

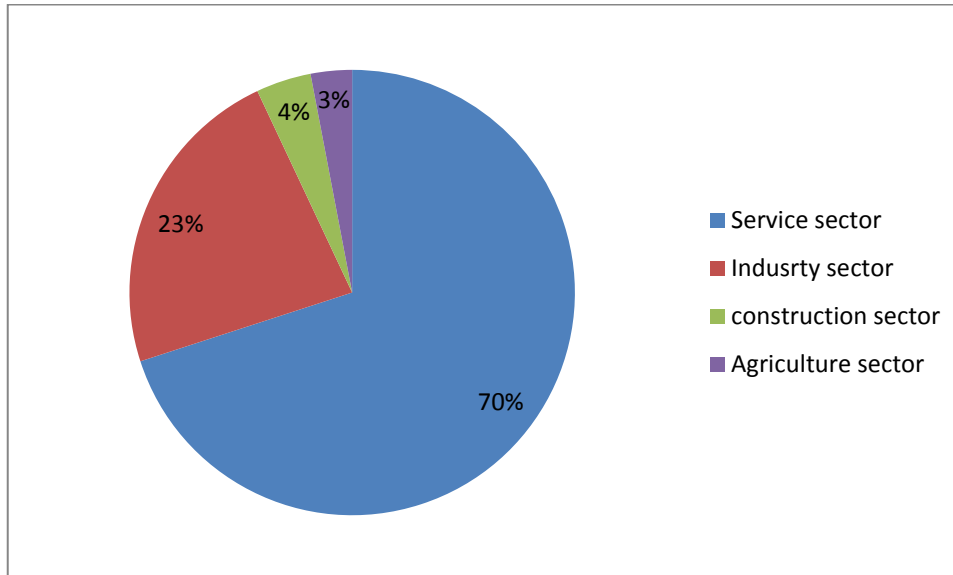
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market Analysis

Project Description

The manufacturing of olive oil soap, using traditional techniques, is one of the oldest industries in Jordan. Despite the availability of other types of soap, demand for olive oil soap is still high in the region and in Arab countries. Olive oil soap has numerous benefits for the skin. The idea was adopted due to the high demand in Jordan and other countries, in addition to the availability of needed raw materials and national experience in this field.

This type of soap can be promoted for daily use and for therapeutic purposes due to its benefits. Famous for olive oil production, Jarash is proposed as the location for setting up the project. The expected amount of production is 250 kg of soap per day in the first year.

Project objectives

The project objectives are summarized in the following points:

- Providing high quality olive oil soap.
- Reduce the volume of imported soap.
- The opportunity for exporting, especially to USA, Europe, Arabian markets.
- Providing job opportunities, due to the experience in this industry.
- Achieving high profit from this project.

Suggested Product

The product is classified under code (340120000). Soap is manufactured in different sizes using olive oil to cover the demand. The product will come in modern designs and shapes to attract customers and will have an average weight of 75 grams per piece.

Target segments

The project targets the following segments:

- Local market.
- Nearby countries.
- USA and Europe, by taking advantage of trade agreements with these countries.

Local Market Analysis

The market size depends on the population of Jordan. Based on the latest census conducted by the Department of Statistics, the population of Jordan is 9,500,000. The export market is expected to include nearby countries, USA, and Europe.

The following table shows the volume of imported and exported soap.

Table 3 volume of imported and exported soap in 2015

Type	Imports (JD)	Imports (kg)	Exports (JD)	Exports (kg)	Re- exported (kg)	Re- exported (JD)	total exports (JD)	Trade balance (JD)
Different types of soap	6,450,558	5,567,667	6,723,130	7,007,511	225,123	216,966	6,948,253	497,695

Department of Statistics/ Economic statistics/ 2015

1.1 Competitor Analysis

The direct competitors for the suggested product are factories that manufacture soap with olive oil or mineral salts. Traditional soap producers are considered indirect competitors.

Direct competitors are:

- Al-Jamal Soap: it contains olive oil.
- Nabulsi soap: the main ingredient is olive oil.
- Bayt Al Saboun: also consists of olive oil and natural plants from Ajloun

Indirect competitors: other makers of traditional soaps.

Project Capacity

The following table shows the expected production capacity:

Table 4 production capacity analysis

Production Capacity Analysis	
Total targeted production in units for the local market	728,000
Total targeted production in units for the outside market	364,000
Total	1,092,000

According to the study, the production capacity will be 3,000 units per day. Around 30% of which for external markets.

The expected daily production rate is based on the assumption that 1,500 units would be produced for the domestic market (75% of total production capacity), while 250 units (25%) for the outside market.

Price Analysis and Pricing Strategies

the suggested price of the product is JD 0.55 /piece in the local market and JD 1.0 /piece for the foreign market. The suggested price is considered to be less than the average price of competitors.

The following table shows the expected price for the local and foreign markets for the next 10 years:

Table 5 expected selling prices during (2018-2027)

Price Estimates	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Price/piece in the local market	0.55	0.57	0.59	0.62	0.64	0.67	0.70	0.72	0.75	0.78
Price/piece in the foreign market	1.00	1.00	1.04	1.08	1.12	1.17	1.22	1.27	1.32	1.37

Expected Revenues Analysis

Revenues in the first year were based on a price of JD 0.55 per piece for the local market /piece and JD 1.0/piece for the foreign market, and depending that 1,500 soap are produced daily for the local market, and 250 soap are produced daily for exportation market.

The following table shows the expected revenues during the period (2018-2027)

Table 6 expected annual revenues for 10 years (2018-2027)

Years	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Price/piece in the local market	0.55	0.57	0.59	0.62	0.64	0.67	0.70	0.72	0.75	0.78
Price/piece in the foreign	1.00	1.00	1.04	1.08	1.12	1.17	1.22	1.27	1.32	1.37
Local market- total production	546,000	546,000	546,000	546,000	546,000	546,000	546,000	546,000	546,000	546,000
Foreign market- total production	91,000	91,000	91,000	91,000	91,000	91,000	91,000	91,000	91,000	91,000
Total production	637,000	637,000	637,000	637,000	637,000	637,000	637,000	637,000	637,000	637,000
Local market revenues	300300	312312	324804	337797	351309	365361	379975	395174	410981	427421
Foreign market revenues	91000	91000	94640	98426	102363	106457	110715	115144	119750	124540
Total revenues	391300	403312	419444	436222	453671	471818	490691	510318	530731	551960

4. Technical Analysis

Required Human Resources

Table 7 required human resources for the project

Job title	number
Factory manger	1
Marketing officer	1
Sales representative	2
Office manager	1
Accountant	1
warehouse keeper	1
Driver	2
Security Guard	1
Number of indirect employees	10
Operators (production line)	3
Maintenance worker	1
Cleaner	1
Number of direct employees	5
Total	15

The following summarize the job descriptions for the required human resources:

- Factory manager: propose the general policy of the factory, prepare special plans and follow up their implementation, supervision on technical and administrative staff, and develop procedures and ensure the achievement of the occupational health and safety requirments.
- Accountant: checking financial subjects in addition to financial auditing, and submit the annual report of the financial situation.

- Marketing officer: examines the needs of the market, follows up the activities of advertising and media, assesses its effectiveness, and monitors the level of sales.
- Office Manager: Organizing and managing the office of the factory manager and assisting him in daily business and ensuring the implementation of the system, in addition to reviewing and approving the schedule of meetings and interviews of the manager, and preparing a weekly plan for the manager.
- Warehouse keeper: Execution of plans and organizing all things related to the stores (raw materials, final product, product under operation and storage areas) and providing reports to the manager and organize the entry and exit of the stock according to the approved procedures.
- Maintenance worker: provide electrical and mechanical maintenance procedures and activities, and follows up the implementation of the planned plans and programs so as to ensure the continuity of the production process and its high quality and with the lowest possible cost.
- Security guard: Implement multiple security instructions to save the factory security system from any threat.

Location Analysis

Jerash is an ideal location for the establishment of the proposed project. Jerash is one of the richest agricultural areas in Jordan, where many crops are grown, especially olive, which is used in the production of the finest olive oil, which is squeezed locally by many traditional and modern contemporary. In Jerash there are the oldest olive trees in the region, some of which extend for hundreds of years, locally its called Roman olive trees, in relation to the era in which Jerash was a Roman city.

Production Process

The idea of the project is to build an olive oil soap factory that has up-to-date equipment and machines for producing high quality products with modern designs and attractive packaging.

Soap is made using a special cooker. Olive oil is heated and then caustic soda solution is added to complete the saponification process. Following that, active ingredients, perfumes and colors are

added. After cooling and kneading the mixture, soap is molded into bars, cut, stamped, packaged and packed in card boxes.

Technical Requirements Analysis

1.1.1 The land and construction works

The project requires 4,000 square meters.

Table8 required land for the project

The land	Area (m ²)
Required land	4,000

The following table shows needed construction works for the project;

Table 9 required constuction works

Construction works	Area (m ²)
Building works	1,000
Designs	1,000
Digging	1,000

Machinery and Equipment

The following table shows the required machines and equipment:

Table 10 required machinery and equipment

Required Machines and Equipment	Qty
Mixing, cutting, molding, stamping machine	1
Mill	1
Cooling unit	1
Mixer	1

Soap Blouder	1
Boiler	1
Laboratory devices	1
Computers	5

Furniture

The factory requires full office furniture for the management department and simple furniture for the factory.

the following table shows required furniture:

Table 11 required furniture for the project

required furniture	Qty
furniture	-

Vehicles and transportation

The following table shows required vehicles for the project:

Table 12 required vehicles

Required vehicles	Qty
Medium Vehicle	2

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.88% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 8 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 13 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 14 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 15 working time assumptions

Working time	
No. of weeks	52
No. of days per week	7
No. of days annually	364
No. of working hours	8

Table 16 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	4.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 17 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	35.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	1.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.50%

Table 18 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	60%
Income Tax after Deduction	8%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 19 Risk Premuim

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	10.12%
Income Tax Rate	8.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 20 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%8
Bets	1.00
Equity	238,528
Loans Value	357,792
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%8.72
Loans Value	357,792
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	238,528
Equity Percentage	%40
Gross	
Project Value	596,320
Weighted Average Cost of Capital	%11.88

Capital Expenditures

The estimated cost of the project is 596,320 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 21 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost(JD)
Required Land	7	4,000	28,000
Total Cost			28,000

Table 22 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross / Cost
Building works	275	1,000	275,000
Design and blueprints	7	1,000	7,000
digging	5	1,000	5,000
Soil test	-	-	200
Other works			14,360
Total			301,560

Table 23 Machinery and equipment Capital Expenditure

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Mixing, cutting, molding, stamping machine	1	20,000	20,000
Mill	1	10,000	10,000
Cooling unit	1	13,000	13,000
Mixer	1	10,000	10,000
Soap Boulder	1	8,000	8,000
Boiler	1	10,000	10,000
Laboratory devices	1	5,000	5,000
Computers	5	300	1,500
Total			77,500

Table 24 Furniture Capital Expenditure

Furniture	Number	Unit cost (JD)	Total cost (JD)
General furniture	-	-	7,000
Total			7,000

Table 25 Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Medium vehicle	2	23,400	46,800
Total cost			46,800

Table 26 Pre-Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	3,000
Transportation	750
Marketing	2,500
Training	2,500
Total cost	8,750

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 27 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	50,374
General and administrative expenses	3	%25	2,036
Marketing expenses	3	%25	489
Indirect Salaries	3	%25	19,600
Total			72,499

Table 28 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	28,000	10%	30,800	%5.2
Construction works	301,560	10%	331,716	%55.6
Machinery & Equipment	77,500	10%	85,250	%14.3
Furniture	7,000	10%	7,700	%1.3
Vehicles	46,800	10%	51,480	%8.6
Pre-Operating Expenses	8,750	10%	9,625	%1.6
Working capital	72,499	10%	79,749	%13.4
Total (JD)	542,109		596,320	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (596,320) JD, where 60% of the project will be financed through loans (357,792) JD, and the remaining (40%) through (Equity) with (238,528) JD. The following table summarizes the general structure of the required financing:

Table 29 Sources of Funding

Funding Sources	Amount	Percentage
Equity	238,528	%40
Loans	357,792	%60
Total	596,320	%100
Use of Fund	Amount	Percentage
Capital Expenditures	506,946	%85.01
Pre-Operating Expenses	9,625	%1.61
Working Capital	79,749	%13.37
Total	596,320	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 30 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	136,955	141,159	146,806	152,678	158,785	165,136	171,742	178,611	185,756	193,186
Service benefit expenses from revenues	19,565	20,166	20,972	21,811	22,684	23,591	24,535	25,516	26,537	27,598
Maintenance from Revenues	5,870	6,050	6,292	6,543	6,805	7,077	7,360	7,655	7,961	8,279
Disposables from Revenues	3,913	4,033	4,194	4,362	4,537	4,718	4,907	5,103	5,307	5,520
Direct Salaries	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Others	9,595	9,903	10,300	10,714	11,144	11,591	12,056	12,540	13,043	13,567
Total	201,498	207,960	216,306	224,988	234,018	243,410	253,179	263,341	273,910	284,903

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 31 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	250	258	265	273	281	290	299	307	317	326
Stationary and Printing	250	258	265	273	281	290	299	307	317	326
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Training	700	721	743	765	788	811	836	861	887	913
Insurance	3,802	3,916	4,034	4,155	4,279	4,408	4,540	4,676	4,816	4,961
Miscellaneous	740	762	785	809	833	858	884	910	938	966
Total	8,142	8,387	8,638	8,897	9,164	9,439	9,722	10,014	10,314	10,624

Marketing Expenses

Table 32 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	1,957	2,017	2,097	2,181	2,268	2,359	2,453	2,552	2,654	2,760

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance, etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 33 expected number of employees

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Factory manger	1	1	1	1	1	1	1	1	1	1
Marketing officer	1	1	1	1	1	1	1	1	1	1
Sales representative	2	2	2	2	2	2	2	2	2	2
Director of the Office	1	1	1	1	1	1	1	1	1	1
Accountant	1	1	1	1	1	1	1	1	1	1
warehouse keeper	1	1	1	1	1	1	1	1	1	1
Driver	2	2	2	2	2	2	2	2	2	2
Security gaurd	1	1	1	1	1	1	1	1	1	1
Total number of indirect employees	10	10	10	10	10	10	10	10	10	10
direct employees										
Operators (production line)	3	3	3	3	3	3	3	3	3	3
Maintenance worker	1	1	1	1	1	1	1	1	1	1

Medical olive oil soap factory

cleaner	1	1	1	1	1	1	1	1	1	1
Total number of direct employees	5	5	5	5	5	5	5	5	5	5
Grand total	15	15	15	15	15	15	15	15	15	15

Table 34 Expected monthly Salaries and Wages (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Factory manger	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Marketing officer	600	625	650	677	705	734	764	795	827	861
Sales representative	350	364	379	395	411	428	445	464	483	502
Director of the Office	400	416	433	451	470	489	509	530	552	574
Accountant	450	468	488	508	528	550	573	596	621	646
warehouse keeper	350	364	379	395	411	428	445	464	483	502
Driver	300	312	325	338	352	367	382	397	414	431
Security gaurd	300	312	325	338	352	367	382	397	414	431
direct employees										
Operators (production line)	350	364	379	395	411	428	445	464	483	502
Maintenance worker	350	364	379	395	411	428	445	464	483	502
cleaner	200	208	217	226	235	245	255	265	276	287

Table 35 Expected Annual Salaries and Wages (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Factory manger	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Marketing officer	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Sales representative	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Director of the Office	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Accountant	7,200	7,495	7,803	8,122	8,455	8,802	9,163	9,539	9,930	10,337
warehouse keeper	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Driver	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Security gaurd	4,800	4,997	5,202	5,415	5,637	5,868	6,109	6,359	6,620	6,891
Total indirect salaries	78,400	81,614	84,961	88,444	92,070	95,845	99,775	103,865	108,124	112,557
direct employees										
Operators (production line)	16,800	17,489	18,206	18,952	19,729	20,538	21,380	22,257	23,169	24,119
Maintenance worker	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
cleaner	3,200	3,331	3,468	3,610	3,758	3,912	4,072	4,239	4,413	4,594
Total direct salaries	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Grand total	104,000	108,264	112,703	117,324	122,134	127,141	132,354	137,781	143,430	149,310

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 36 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	30,800	%0.0	%0.0
Construction Works	331,716	%5.0	%0.0
Machinaries	85,250	%10.0	%2.0
Furniture	7,700	%10.0	%5.0
Vehicles	51,480	%10.0	%0.0
	506,946		

Table 37 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	506,946	509,036	511,179	513,378	515,633	517,946	520,320	522,756	525,256	527,823
Total Depreciation Expenses	31,029	31,238	31,452	31,672	31,897	32,129	32,366	32,610	32,860	33,116
Total Accumulated Depreciation	31,029	62,267	93,719	125,391	157,288	189,417	221,783	254,393	287,253	320,369
Total Additions	0	2,090	2,143	2,198	2,255	2,314	2,374	2,436	2,500	2,566
Total Net Book Values	475,917	446,769	417,461	387,987	358,345	328,529	298,537	268,363	238,004	207,454

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 38 Loan Details

Loan Value		357,792			
Annual Interest Rate		9.48%			
Loans Period		5			
Grace Period		0			
Loan Starts at year:		2018			
Annual Payment		93,134			
Number of Payments		5			
Year	Annual Payment	Interest	Capital	Loan Remaining Value	
2018				357,792	
2019	93,134	33,919	59,215	298,577	
2020	93,134	28,305	64,829	233,748	
2021	93,134	22,159	70,975	162,773	
2022	93,134	15,431	77,703	85,070	
2023	93,134	8,065	85,070	0	

Income Statement

Table 39 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	391,300	403,312	419,444	436,222	453,671	471,818	490,691	510,318	530,731	551,960
Gross Operating Revenues	391,300	403,312	419,444	436,222	453,671	471,818	490,691	510,318	530,731	551,960
Operating Expenses	(201,498)	(207,960)	(216,306)	(224,988)	(234,018)	(243,410)	(253,179)	(263,341)	(273,910)	(284,903)
Gross Operating Profit	189,802	195,352	203,138	211,234	219,653	228,408	237,512	246,978	256,821	267,057
<i>Gross Profit Percentage</i>	<i>%49</i>	<i>%48</i>	<i>%48</i>	<i>%48</i>	<i>%48</i>	<i>%48</i>	<i>%48</i>	<i>%48</i>	<i>%48</i>	<i>%48</i>
Salaries and Benefits (Indirect	(78,400)	(81,614)	(84,961)	(88,444)	(92,070)	(95,845)	(99,775)	(103,865)	(108,124)	(112,557)
General and Administraive	(8,142)	(8,387)	(8,638)	(8,897)	(9,164)	(9,439)	(9,722)	(10,014)	(10,314)	(10,624)
Markeing Expenses	(1,957)	(2,017)	(2,097)	(2,181)	(2,268)	(2,359)	(2,453)	(2,552)	(2,654)	(2,760)
Pre-Operating Expenses	(9,625)									
Gross Indirect Expenses	(98,124)	(92,018)	(95,696)	(99,522)	(103,503)	(107,643)	(111,950)	(116,431)	(121,092)	(125,941)
Income before Interest,	91,679	103,334	107,442	111,712	116,151	120,765	125,561	130,547	135,729	141,116
Fixed Assets Depreciations	(31,029)	(31,238)	(31,452)	(31,672)	(31,897)	(32,129)	(32,366)	(32,610)	(32,860)	(33,116)
Income before Tax and Interests	60,650	72,097	75,990	80,040	84,253	88,636	93,195	97,937	102,869	108,000
Bank Interests	(33,919)	(28,305)	(22,159)	(15,431)	(8,065)	(0)	(0)	(0)	(0)	(0)
Income Before Tax	26,731	43,792	53,831	64,609	76,189	88,636	93,195	97,937	102,869	108,000
Income Tax	(2,138)	(3,503)	(4,306)	(5,169)	(6,095)	(7,091)	(7,456)	(7,835)	(8,230)	(8,640)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	24,593	40,288	49,524	59,440	70,094	81,545	85,739	90,102	94,640	99,360
<i>Net Profit Percentage</i>	<i>%6</i>	<i>%10</i>	<i>%12</i>	<i>%14</i>	<i>%15</i>	<i>%17</i>	<i>%17</i>	<i>%18</i>	<i>%18</i>	<i>%18</i>
Compulsory Reserves	(2,459)	(4,029)	(4,952)	(5,944)	(7,009)	(8,155)	(8,574)	(9,010)	(9,464)	(9,936)
Retained Earnings	22,133	58,393	102,965	156,461	219,545	292,936	370,101	451,193	536,369	625,792

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 40 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	24,593	40,288	49,524	59,440	70,094	81,545	85,739	90,102	94,640	99,360
Bank Interests										
Depreciation	33,919	28,305	22,159	15,431	8,065	0	0	0	0	0
Total Operating Cashflows before Additions to Working Capital	31,029	31,238	31,452	31,672	31,897	32,129	32,366	32,610	32,860	33,116
Inventory (Increase/Decrease)	(11,413)	(350)	(471)	(489)	(509)	(529)	(550)	(572)	(595)	(619)
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0

Medical olive oil soap factory

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Accounts Payables (Increase/Decrease)	10,757	376	450	468	487	507	527	549	571	594
Working Capital (Increase/Decrease)	(656)	26	(21)	(21)	(22)	(22)	(23)	(24)	(25)	(25)
Net Cashflows from Operating Activities	88,884	99,857	103,115	106,522	110,034	113,651	118,082	122,688	127,475	132,451
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(506,946)	(2,090)	(2,143)	(2,198)	(2,255)	(2,314)	(2,374)	(2,436)	(2,500)	(2,566)
Net Cashflows from Investments Activities	(506,946)	(2,090)	(2,143)	(2,198)	(2,255)	(2,314)	(2,374)	(2,436)	(2,500)	(2,566)
Cashflows from Financing Activities										
Capital	238,528									
Loan Amortization	(59,215)	(64,829)	(70,975)	(77,703)	(85,070)	0	0	0	0	0
Bank Interest Rate	(33,919)	(28,305)	(22,159)	(15,431)	(8,065)	(0)	(0)	(0)	(0)	(0)
Loans	357,792	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	503,186	(93,134)	(93,134)	(93,134)	(93,134)	0	0	0	0	0
Net (Increase/Decrease) in Cash	85,124	4,633	7,838	11,190	14,645	111,338	115,708	120,252	124,975	129,885
Cashflows at the Beginning of Period	0	85,124	89,757	97,595	108,785	123,429	234,767	350,476	470,727	595,702
Cashflows at the End of Period	85,124	89,757	97,595	108,785	123,429	234,767	350,476	470,727	595,702	725,587

Expected Balance Sheet

Medical olive oil soap factory

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 41 Expected balance sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	85,124	89,757	97,595	108,785	123,429	234,767	350,476	470,727	595,702	725,587
Inventory	11,413	11,763	12,234	12,723	13,232	13,761	14,312	14,884	15,480	16,099
Total current Assets	96,537	101,520	109,829	121,508	136,661	248,528	364,787	485,612	611,182	741,686
Non Current Assets										
Fixed Assets (net)	475,917	446,769	417,461	387,987	358,345	328,529	298,537	268,363	238,004	207,454
Total Non Current Assets	475,917	446,769	417,461	387,987	358,345	328,529	298,537	268,363	238,004	207,454
Total Assets	572,454	548,290	527,289	509,495	495,006	577,058	663,324	753,975	849,185	949,139
Liabilities										
Current Liabilities										
Paybles	10,757	11,133	11,583	12,052	12,539	13,046	13,573	14,122	14,692	15,286
Remaining amount of Loan	64,829	70,975	77,703	85,070	(0)	(0)	(0)	(0)	(0)	0
Total current Liabilities	75,586	82,108	89,287	97,121	12,539	13,046	13,573	14,122	14,692	15,286
Non Current Liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Long Terms Loans	233,748	162,773	85,070	0	0	0	0	0	0	0
Total Long Term Liabilities	233,748	162,773	85,070	0	0	0	0	0	0	0
Total Liabilities	309,334	244,881	174,356	97,121	12,539	13,046	13,573	14,122	14,692	15,286
Owners Equity										
Shareholders Contributions	238,528	238,528	238,528	238,528	238,528	238,528	238,528	238,528	238,528	238,528
Statutory Reseve	2,459	6,488	11,441	17,385	24,394	32,548	41,122	50,133	59,597	69,532
Retained Profits	22,133	58,393	102,965	156,461	219,545	292,936	370,101	451,193	536,369	625,792
Total Equity	263,121	303,409	352,933	412,373	482,467	564,012	649,751	739,853	834,493	933,853
Total Liabilities and Equity	572,454	548,290	527,289	509,495	495,006	577,058	663,324	753,975	849,185	949,139

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans; therefore, the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 42 Free Net Cash Flows

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
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Net Free Cashflows	(596,320)	98,509	97,767	100,972	104,324	107,779	111,338	115,708	120,252	124,975	129,885	1,714,004
Discount Factor	1.00	0.71	0.64	0.57	0.51	0.46	0.41	0.36	0.33	0.29	0.26	0.26
Net Present Value for Cash Flows	(596,320)	70,341	62,398	57,600	53,192	49,118	45,352	42,127	39,132	36,350	33,766	445,592

The table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 43 Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(596,320)	98,509	97,767	100,972	104,324	107,779	111,338	115,708	120,252	124,975	129,885
Project Value Returned	596,320	497,811	400,044	299,072	194,748	86,969	(24,369)	(140,077)	(260,329)	(385,304)	(515,188)
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 44 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.88
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Net Present Value for Cashflows	338,647
Payback Period	5
Internal Rate of Return	%22.34

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 45 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%22.34	5	11.88%
Revenues declined by 10%	%18.15	7	11.88%
Operating Expenses Increased by 10%	%18.27	7	11.88%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 338,647 Jordan Dinars considering that the project provides 15 Job Opportunities for the governorate residents.

